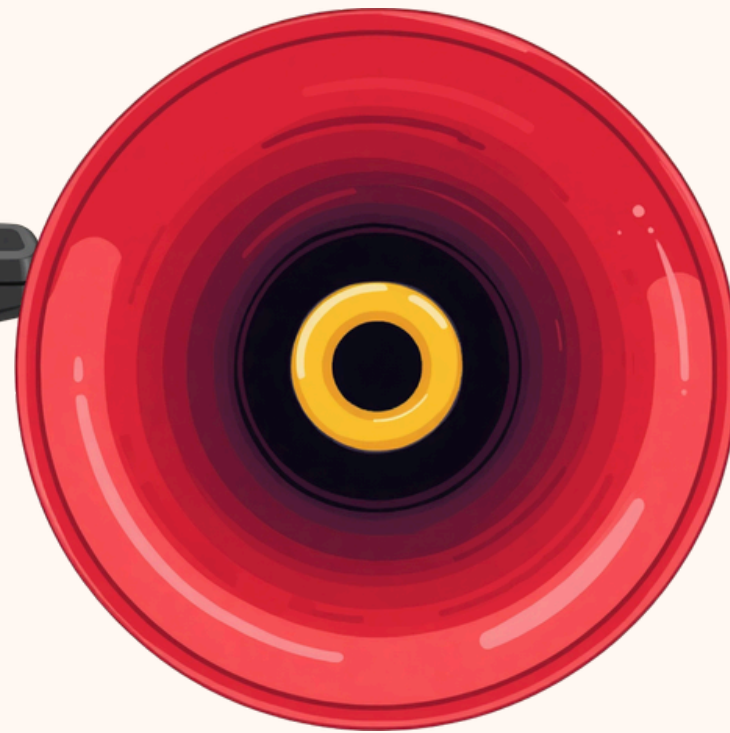




HARDWARE CAPSTONE: DESIGN A USEFUL GADGET

SPOT THE GADGET!

Which ones are gadgets? What problems do they solve?





PRESSURE COOKER

Cooks food faster using
steam pressure – saves time
and fuel!



WHAT IS A GADGET?

PENCIL BOX

Holds fabric together safely – no
more falling badges!



SAFETY PIN

Keeps pencils, erasers and
sharpeners organised in
compartments!



GADGETS SOLVE PROBLEMS

IDLI STAND

Stacks steaming plates to cook many idlis at once without taking extra space!

CYCLE PUMP

Keeps homemade food warm while delivering it across the city!

DABBA DELIVERY SYSTEM

Fills bicycle tyres easily so you can ride smoothly anywhere!





MATCH THE PROBLEM TO THE GADGET



SLOW COOKING TAKES TOO
LONG

Draw a line to match each
problem on the left with the
gadget that solves it!

PRESSURE COOKER

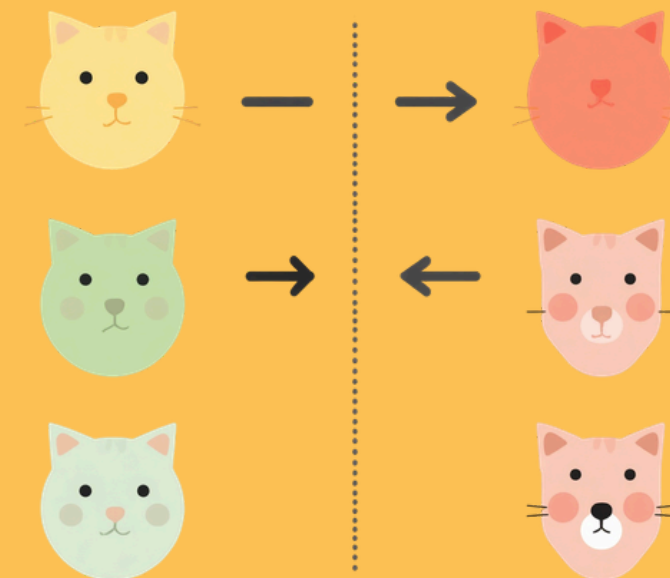
Safety Pin

BADGE KEEPS FALLING OFF
CLOTHES

Tiffin Box

#Matching activity

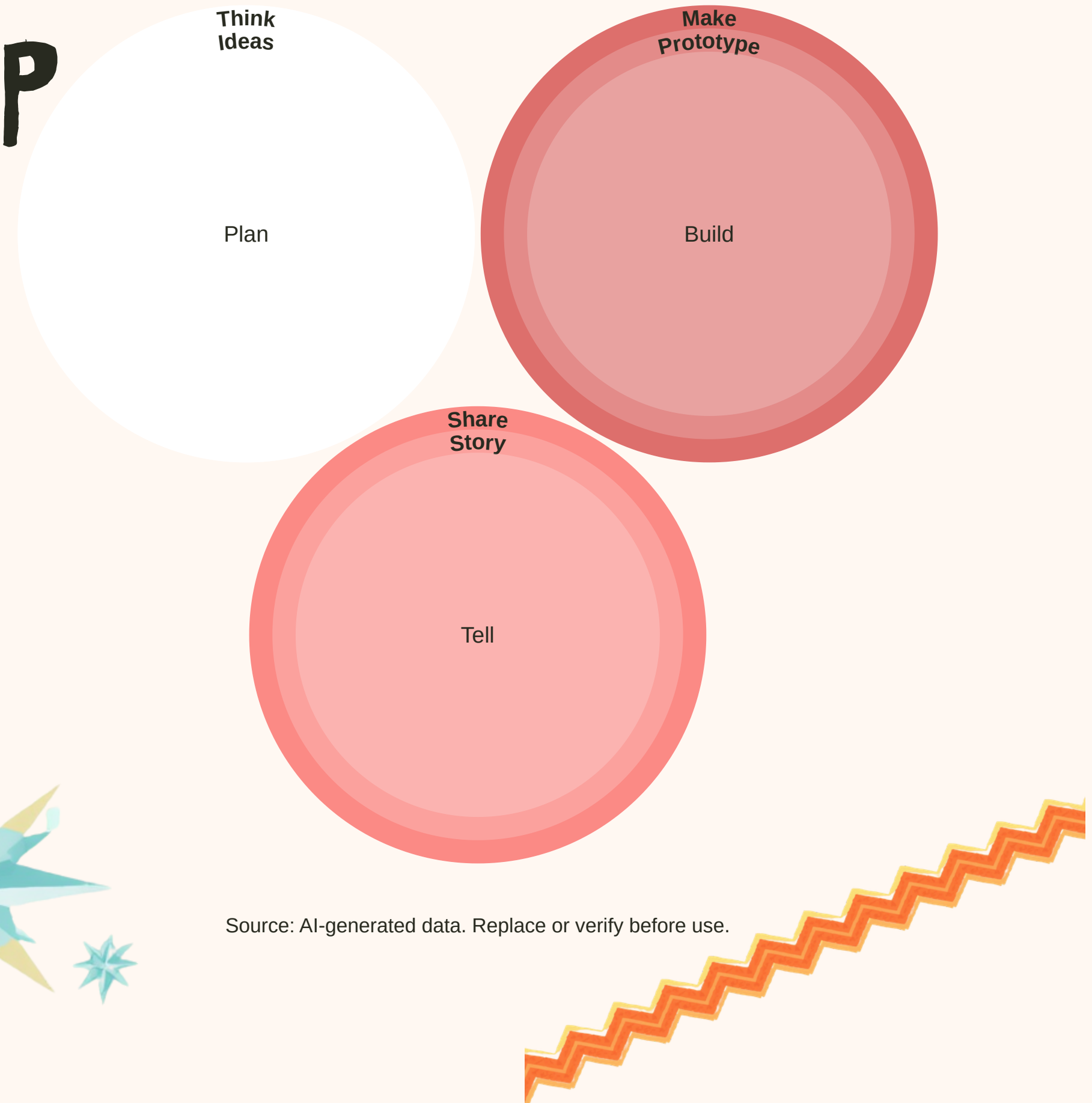
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THE DESIGN LOOP

- Design means THINK about problems, MAKE gadgets, and SHARE your idea.
- This loop helps you create amazing gadgets that solve real problems!

- THINK → MAKE → SHARE
- Keep looping to improve your design!



Source: AI-generated data. Replace or verify before use.

THE LEAKY WATER BOTTLE PROBLEM

RAVI'S PROBLEM

Ravi's water bottle leaks and wastes water every day at school. The water drips into his bag and makes his books wet!

HOW CAN HE FIX IT?

Can you think of ways Ravi could fix his leaky bottle? What materials might help?



MANSUKHBHAI PRAJAPATI'S MITTICOOL CLAY FRIDGE

AN AMAZING INDIAN INNOVATOR

Mansukhbhai made a fridge from clay to keep food cool without electricity! He noticed that many families in villages had no power, so he created the Mitticool fridge using traditional pottery skills. Now families can keep their vegetables and milk fresh!





YOUNG AND LOCAL INNOVATORS

REMYA JOSE

A Kerala girl who invented a human-powered washing machine to help her mother!



HONEYBEE NETWORK

A network that collects clever grassroots ideas from villages across India!

ATL STUDENTS

School students creating amazing inventions in Atal Tinkering Labs!



YOUR TURN: MY PROBLEM STATEMENT

FILL YOUR WORKSHEET

What problem do you want to solve? Think about something that bothers you or someone you know.

WORKSHEET FIELDS TO COMPLETE

- I noticed a problem when...
- The problem is...
- This bothers...
- I think a gadget could help because...





PROBLEMS YOU CAN SOLVE



READING TROUBLE

Grandmother can't read
small print on medicine
bottles.



HEAVY BAGS

Heavy school bags hurt
when climbing stairs.

WASTING WATER

School water tap drips and wastes
water every day.



THINK ABOUT YOUR PROBLEM

- Why does this problem matter? Who does it affect?
- Before you start building, take a moment to really understand your problem. Think about all the people who face this problem every day.
- Good designers always think deeply before they make!



NEXT STEP: MAKE YOUR GADGET

GET READY TO BUILD!

Soon, you will build your gadget to solve your problem. Use simple materials like cardboard, tape, string, and recycled items to bring your idea to life!



THEN, SHARE YOUR IDEA

- You will tell others how your gadget helps.
- Sharing is the final step of the Design Loop. When you share, you explain your idea and show how it solves a problem!
- Great designers share their ideas with others to help more people.



3-2-1 EXIT



3 GADGETS

Name 3 gadgets you
learned about today!

1 FEELING WORD

Which 2 steps of THINK →
MAKE → SHARE do you
remember?

2 DESIGN LOOP STEPS

Share 1 word about how you feel
after today's session!





WHAT'S NEXT?

SESSION 2: EXPLORE MATERIALS

Learn about different materials and tools you can use to build your gadget!

SESSION 3: BUILD YOUR PROTOTYPE

Create your first gadget model using the materials you explored!

SESSION 4: TEST & IMPROVE

Make changes to your design based on what works and what doesn't!



TAKE-HOME CHALLENGE

- Look around your home or school. Find one problem you want to solve!
- Notice things that don't work well or could be better. Maybe a door that squeaks, a shelf that's too high, or something that spills easily.
- You are a problem-finder now! Designers see problems everywhere.





THANK YOU FOR
JOINING!
YOU ARE ALREADY A
DESIGNER!



QUESTIONS & READY TO START!

Any questions? Let's get ready to design!

